A Telemetry-based PI Tuning Strategy for Low-level Control of an Omnidirectional Mobile Robot*

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Abstract. Mobile robot control requires precision and accuracy to react to unpredictable situations. However, it is non-trivial to design a low-level controller for each wheel that combines fast response and stability in the presence of disturbances. The Proportional-Integral (PI) controller algorithm is the most used technique for DC motors; however, tuning these controllers can become a time-consuming task depending on the number of robots and tuning methodology. We propose a novel telemetry-based strategy to find suitable PI controller parameters more quickly using accurate motor models of the omnidirectional mobile robots obtained from an on-site data sampling mechanism. We evaluate our approach on an omnidirectional robot designed within the RoboCup Small Size League (SSL) competition rules. The results compare the proposed method with a based on Quantitative Feedback Theory (QFT) approach. Our strategy improved on average the robot's performance by 17.95% when using Integral Absolute Error (IAE) and by 12.75% when using Integral Squared Error (ISE) criteria.

Keywords: Low-level Control · PI Tuning · Omnidirectional Mobile Robot

1 Introduction

An Omnidirectional Mobile Robot (OMR) is a type of holonomic robot widely used in dynamic environments with the ability to translate and rotate independently in a two-dimensional plane. The RoboCup Small Size League (SSL) competition is a highly dynamic environment where two teams with up to eleven OMRs each compete within rules adapted from FIFA soccer. Currently, the teams use a shared vision system [22]. However, the league is moving towards an onboard vision for each robot. All groups are encouraged to develop their systems through technical challenges. This perspective change requires the teams to design a very reliable low-level control for their robots without external feedback. In the context of robot competitions such as robot soccer, a robot must

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act fast and precisely because defenders of the opposing team quickly block the opportunities to score a goal.

A significant number of works have investigated the control of OMRs [1, 7, 10]. The leading causes of trajectory tracking errors are nonlinearities due to slippage between wheel and ground, friction, vibrations, and disturbances [16]. To reduce these errors, we have to design an accurate motor control system for each OMR wheel. One of the most used approaches in the literature to design, analyze, verify, and validate dynamic systems is the model-based design [15]. Model-based design techniques use approximate mathematical models of the system to estimate unknown parameters, such as Proportional, Integral, and Derivative (PID) controller parameters for DC motor control [12]. Linearization and discretization techniques can simplify dynamic models and satisfy real-time constraints [10].

Works dealing with the motor model acquisition and PID tuning generally use robots connected to embedded computers communicating through serial communication [1, 6]. The methodology used to identify the motor model differs among existing approaches [6, 12, 21]. The use of data acquired only in simulation for tuning and validation of controllers is widespread in works in the area [4, 16]. When working in a multi-robot environment with robot size restrictions, it can be very time-consuming to tune a PID controller for each robot's motors in a new environment. It usually requires a lot of human effort [9]. It is also essential to establish wireless communication so that the robot can move freely in an environment of data acquisition and real validation.

This work presents a novel telemetry-based PI tuning strategy for low-level control of omnidirectional mobile robot motors. We propose a data-collecting mechanism that allows the PI tuning of the motors using data sampled on-site in a real multi-robot environment with low latency. The proposed strategy includes a technique based on Multi-level Pseudo-Random Sequence (MPRS) of signals for motor model acquisition. The use of this signal improves the linearized motor model estimation accuracy compared to traditional techniques based on Pseudo-Random Binary Sequence (PRBS). The telemetry-based strategy allows a fast tuning of the PI constants for multiple robots without changing the embedded code. The proposed strategy is implemented and validated with a robot built under the RoboCup SSL rules and compared with an approach based on the QFT technique. Our main contributions are:

1. A data sampling engine based on wireless communication supporting on-site data sampling from multiple robots with low latency,
2. A method for system identification using MPRS signals to identify linearized model based on specific motor parameters, and
3. A novel strategy for fast tuning of a PI controller for low-level control in an OMR.

The rest of this paper is organized as follows: Section 2 presents the related works. Section 3 details the system architecture, model acquisition, and PI tuning strategy. Section 4 presents the experimental results. Finally, Section 5 concludes the work and presents further works.

2 Related Works

System identification is the process of constructing mathematical models of the system under study from prior knowledge [17]. In mobile robots, different system identification methods have been applied to derive a DC motor model. The most common is to apply Random Binary Sequences to the motor to collect motor response information and use this data to obtain a transfer function of the motor [13]. Sahputro et al. [20] used Pseudo-Random Binary Sequence (PRBS) to get the estimated transfer function of a DC motor to control a mobile robot using the Recursive Least Square (RLS) method. Hat et al. [12] also applied PRBS signal for determining a brushless DC motor model using MATLAB software. Braun et al. [5] presented an identification methodology using Multi-level Pseudo-Random Sequence (MPRS) applied to a Rapid Thermal Processing (RTP) reactor plant as one of its case studies. Although the MPRS approach was applied primarily with process systems, systems with different model structures can also use this method. One advantage of the MPRS approach is that it can use different values within the motor operating range, while the PRBS uses only two values to perform the identification.

Data sampling and acquisition are crucial steps to perform reasonable control of mobile robots. Recent works present embedded computers using serial communication to read motor data [6, 21]. When working with mobile robots with tight constraints such as size or cost limitations, there is a need to use lower-cost embedded development boards. An alternative is to use wireless communication to send and receive information from a desktop that can compute complex algorithms without those restrictions.

Different approaches have been proposed to address the low-level control for four-wheeled OMRs. The approach described in [11] used a discrete-time Linear Quadratic Tracking approach (LQT) as a low-level controller for the trajectory tracking of an OMR. Al et al. [1] improved the work in [11] presenting an embedded system for motion control. The system controls an OMR using Linear Quadratic Regulator (LQR) as a low-level controller. A fuzzy-PI controller controls the position of the OMR as a high-level controller for solving the problem of trajectory tracking. Comasolivas et al. [6] presented a control technique called Quantitative Feedback Theory (QFT) for low-level control for an OMR. The QFT technique is a frequency domain technique based on the Nichols chart, offering a robust design over specified frequency domain tolerances. The QFT approach identifies multiple linear models to design a PI controller in a standard platform called iSRob, comparing its experimental results with a gain-scheduling PI-controller based on pole assignment.

Most of the works that address the problem of trajectory tracking require an external source such as a vision system to be used as position feedback instead of robot odometry due to the use of a high-level controller [1, 11]. When working in a dynamic multi-robot environment, the approach proposed by [6] can be time-consuming because it estimates multiple linear models, and the loop shaping controller stage requires manual adjustments for each controller.

3 Telemetry-based PI Tuning Strategy

The proposed strategy for designing the low-level controller has four main stages: motor parameters acquisition, motor model estimation, PI tuning, and validation. For the motor parameters acquisition stage, we defined the encoder sampling rate for the motors. We also developed a data sampling engine for finding the motor's rise time and settling time using data sampled from the motors attached to a real SSL robot moving with all loads acting on it in a competition field.

In the motor model estimation stage, we propose using a Multi-level Pseudo-Random Sequence (MPRS) as the signal input of the motors. The generated MPRS signals are sent to the robot through wireless communication using the mentioned data sampling engine. For each signal received, the robot returns a message containing the four motors' speed (in rad/s). Then, this collected data is used to find the linearized motor model estimation following the methodology proposed by [12].

The PI controller's tuning stage can find suitable control constants for the application using the motor's approximate model for instant validation. We use the identified model from the model estimation stage as input to tuning a PI controller based on the design requirements. The control constants found are sent to the robot through telemetry for validation in the next step.

The validation stage consists of implementing the discrete-time PI controller with tuned parameters as a low-level controller in a real robot. We validate by sending 20 different signal levels during a specific time and receive the robot's speed through telemetry. Finally, we use Integral Absolute Error (IAE) and Integral Squared Error (ISE) performance metrics to evaluate the validation results.

3.1 Data Capturing Mechanism

This section will present the proposed strategy for capturing and sampling data from the motors in a robot. Figure 1 shows an overview of the proposed system architecture. The control of each robot includes high-level control and low-level control. The high-level control is implemented in the external computer connected to a base station through Ethernet protocol. This control sends different control messages such as the desired robot's speed, selected motor's PWM, or control constants. The base station includes a NUCLEO-H743ZI2 development board responsible for communicating with the robot using radio modules (nRF24L01+).

The low-level control is implemented in the robot using a NUCLEO-F767ZI development board and communicates with the base station using radio modules (nRF24L01+). The robot decodes the control message received and acts accordingly for each wheel.

This proposed data acquisition engine includes different types of messages that use the presented system architecture to communicate with each robot for

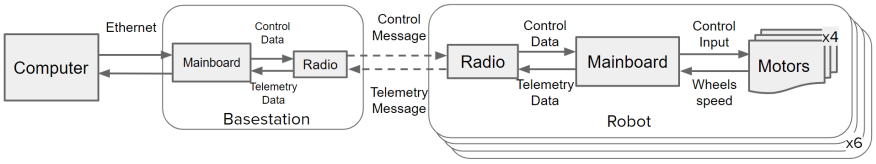


Fig. 1. System architecture overview.

sampling motor data to perform the model estimation, send the PI controller parameters after tuning, and validate the low-level control.

The proposed data collecting mechanism based on wireless communication allows the sample more precise motor data in the real-world environment (on-site). Another advantage of this mechanism is the possibility of configuring tuning parameters online by sending configuration messages to adjust the PI controller constants without changing the embedded code.

3.2 Motor parameters and system identification

The encoder sampling rate is one of the key parameters in a discrete controller. It determines the measurement frequency of the number of pulses read by the sensor, both clockwise and counterclockwise. Therefore, the higher the sampling rate, the faster the motor speed updates. However, there is a trade-off between the sampling rate and the uncertainty of the encoder. The faster the encoder update frequency, the greater the associated uncertainty. Equation 1 shows how to obtain the uncertainty (Δ) for one encoder pulse:

$$\Delta = 2\pi \cdot \frac{1}{EncoderMode \cdot PPR \cdot t_{sample}} \tag{1}$$

The *EncoderMode* refers to the quadrature encoding type, which can be 1, 2, or 4. PPR refers to the number of pulses per revolution, and T_{sample} refers to the time between each encoder update.

This work uses Maxon's Inductive Little Encoder (MILE), designed for the EC-45 flat motors from Maxon Global. When choosing the encoder sampling rate, it is necessary to consider the hardware's ability to process the inner closed-loop control in real-time. To control the motor efficiently, the time between the encoder samples must be at least 5 to 10 times lower than the motor rise time to a step input [8].

To find the motor's rise time and settling time, we use the mentioned data sampling mechanism to send a step signal control message and to receive the motors' response through telemetry messages. The rise time is when it takes the system to go from 10% to 90% of the specified value [19]. The settling time is when it takes the system to be stable at around 2% of the final value [19]. These parameters are essential for determining the design requirements.

The complexity of the driver and motor system makes the formal definition of the motor model mathematically very difficult due to the high number of electrical components involved. To solve that, we can approximate the motor model using its response when applying a signal as input. This work proposes using MPRS signal for system identification as it has an amplitude varied and fixed frequency. A larger range of inputs allows a better estimation of the system's dynamic behavior as it covers a more significant number of input possibilities. One of the main advantages of the MPRS signal is that it can afford more nonlinearities as it does not vary its frequency pulses depending on the application environment. Table 1 shows the parameters for the MPRS signal generation MPRS signals.

Table 1. Parameters used for MPRS signal generation.

Signal	Parameter	Value
MPRS	Number of Samples	511
	Sampling Period	0.3 (sec)
	Input Range	−100 to 100

Figure 2 shows a fragment of the MPRS signal applied to one of the robot motors with its response. The MPRS signals are applied to the motors remotely through specific messages. The motor's responses also received by messages are used to estimate a first-order discrete-time transfer function using an Output-Error (OE) model with the aid of MATLAB's System Identification Toolbox [18]. The OE algorithm relates data measured inputs to outputs. It returns the estimated values of the model parameters with a percentage of how well the model's response fits the input data. The nonlinear least squares with an automatically chosen line search algorithm are used to estimate the percentage of fitting of the identified function. The transfer function estimated allows the design of an initial PI controller to control the motor safely, reducing the risks of damaging the equipment [2].

3.3 Motor Control and PI Tuning

A PI controller is the best controller for this type of the first-order system. Previous works in the literature show that the derivative term does not bring many advantages in this control system by performing a root-locus analysis, and has the drawback of amplifying noise [3, 6]. The method used to discretize the PI controller was the backward difference method [3]. The digital PI controller implemented is described in Equation (2):

$$u(t) = K_p \cdot e(t) + K_i \cdot T_s \cdot \min\left(\sum_{i=0}^{t} e(t), error_{threshold}\right) + u_0 \qquad (2)$$

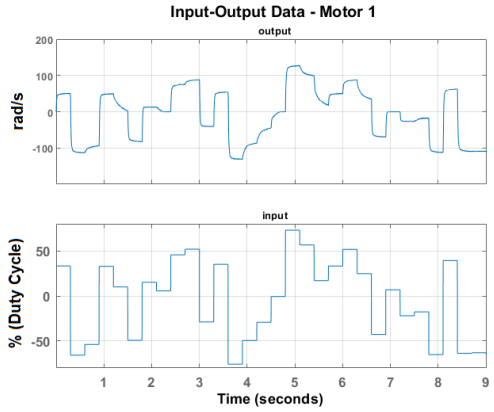


Fig. 2. Fragment with nine seconds of the motor response (in rad/s) to an MPRS input signal (in % Duty Cycle).

The u_0 term is the controller's initial output value. The integral term is the integral constant (K_i) multiplied by the sampling time (T_s) and the sum of all errors during the control execution time. The proportional term is the proportional constant (K_p) multiplied by the actual error ($e(t)$). The backward difference method converts the integral and derivative terms to their corresponding discrete parts over time. An integral anti-windup was implemented to avoid saturation of the integrator term, considering the minimum between the sum of errors and the maximum error threshold ($error_{threshold}$).

The PI tuning step uses the fitted models of the motors to adjust the discrete controller using MATLAB's interactive tool *PID Tuner*. The designer must specify predefined parameters in the tool based on the OMR application: response time and transient behavior. The response time impacts the maximum rise time and maximum settling time of the system, and the transient behavior influences maximum overshoot.

The validation step consists of using the telemetry to measure the response of the robot's motors with the tuned controller. In this validation we sent 20 different speed values during 0.5 seconds each. We evaluate these responses measuring the error of each experiment with two metrics: Integral Absolute Error (IAE) and Integral Squared Error (ISE). Given $e(t)$ the error in the time t. We choose these metrics, based on previous works [6, 14]. Equation (3) and (4) presents how the IAE and ISE criteria is calculated, respectively:

$$IAE = \int_0^\infty |e(t)|dt \qquad (3) \qquad\qquad ISE = \int_0^\infty e(t)^2 dt \qquad (4)$$

4 Case Study and Results

This section presents the experimental results to validate the proposed telemetry-based strategy for the low-level controller. A case study validates the proposed approach with a robot designed for the RoboCup SSL competition. We present the motor parameter acquisition results, compare the proposed MPRS signal for system identification to a PRBS signal, and the proposed PI tuning approach to a robust QFT controller implemented in an SSL robot under the same conditions.

4.1 Motor Parameters

First, to acquire the motor parameters, we need to determine the encoder sampling rate, which needs to be large enough to detect the motor's transient response and identify external noise's influence. Table 2 shows some uncertainty values for each interval between samples based on Equation 1 using a 1024 PPR MILE encoder.

Table 2. Relationship between the time between samples (T_{sample}) in seconds and associated uncertainty (Δ) in RPM and rad/s.

$[T_{sample}](s)$	$[\Delta](RPM)$	$[\Delta](rad/s)$
0.001	14.6484	1.534
0.002	7.3242	0.767
0.005	2.9297	0.307
0.01	1.4648	0.153

As mentioned in Section 3.2, the time between samples must be at least 5 to 10 times lower than the motor rise time to a step input. We applied a step using the telemetry system to measure the motor response, and Table 3 shows the results with 30 repetitions and their respective standard deviation.

Table 3. Step response characteristics for a Maxon EC-45 flat 50W motor with 30 repetitions, the rising time mean ($\overline{T_r}$) and settling time mean $\overline{T_s}$ are in seconds and $\sigma(T_r)$ and $\sigma(T_s)$ represent their standard deviations.

Rising Time		**Settling Time**	
$[\overline{T_r}](s)$	$[\sigma(T_r)]$	$[\overline{T_s}](s)$	$[\sigma(T_s)]$
0.049	0.006	0.125	0.064

With the results in Table 2 and Table 3 we can conclude that the T_{sample} should be lower than 0.01 seconds, depending on the maximum uncertainty defined by the designer in the system requirements.

4.2 System identification for the motors

The motor model identification process provides a transfer function of the system. This transfer function is the motor's digital representation as it simulates its response to signal inputs accurately. In this work, we proposed using an MPRS signal to identify the motor model with high accuracy.

Since the MPRS signal has not been used before to identify DC motors, we compare the identification results using the proposed MPRS signal and a PRBS signal implemented by [20]. MATLAB's System Identification toolbox processes the collected data through the robot telemetry messages from the base station to identify both transfer functions. Table 4 show experimental results representing the percentage of fitting for the identified discrete-time transfer function for the PRBS signal and proposed MPRS signal.

Table 4. Percentage of fitting for the identified discrete-time transfer function with classic PRBS and proposed MPRS signal inputs for each robot's motor.

Motor (#)	PRBS Fitting(%) [20]	MPRS Fitting(%)
1	65.76	91.09
2	65.44	91.22
3	64.9	91.38
4	65.54	91.08

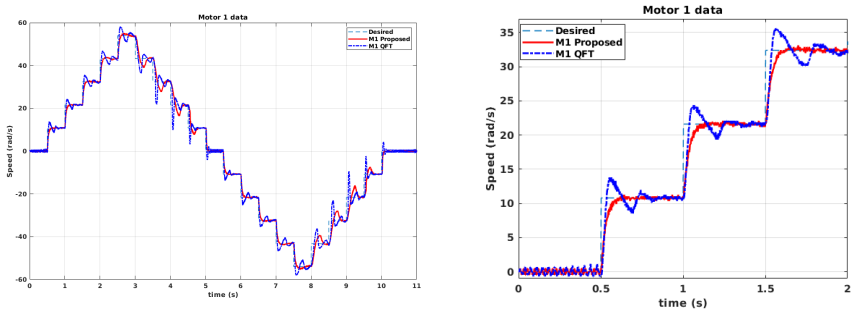
The proposed MPRS identification method improved the PRBS method by an average of 39.41%. This improvement makes the constants obtained in the tuning stage more reliable and directly impacts the experimental results of the low-level controller.

4.3 Motor Control and PI Tuning

The proposed strategy is compared with the control approach called Quantitative Feedback Theory (QFT) using the performance specifications presented by [6]. Figure 3a shows the motor response for one motor in an SSL environment, and Figure 3b presents a detailed view of the motor response for a better transient behavior understanding. The proposed strategy demonstrated a significant reduction in the overshoot, which reduces the error in performance evaluations, avoids current peaks during acceleration, and collaborates for a smoother movement in a global aspect.

Figure 4 presents the error (in rad/s) during time of the proposed strategy and QFT approach for the inputs applied in Figure 3a. The error analysis shows how overshoot impacts performance metrics as the area under the curve for the QFT approach is significantly higher than the area in the proposed strategy.

To compare the proposed approach with the QFT technique, we executed the validation experiments 30 times for each motor due to the non-deterministic



(a) (b)

Fig. 3. Results for the proposed approach and QFT in one robot motor: (a) comparison between reference speed (dashed line) and angular speed with proposed strategy (solid line) and QFT approach (dash-dot line); (b) detailed view of the first two seconds.

aspect. Table 5 shows the average of the IAE and ISE metrics for each technique. The results show that the proposed approach reduced on average the IAE by 17.95% and the ISE by 12.75% when compared to the QFT approach.

Table 5. Average results in a real scenario with the proposed strategy and the QFT approach using IAE and ISE criteria for each motor.

Criteria	Motor	Proposed	QFT [6]
IAE	1	15.92	19.19
	2	16.64	20.29
	3	16.96	20.86
	4	16.84	20.56
ISE	1	92.86	104.40
	2	96.94	111.82
	3	95.84	111.31
	4	97.07	111.26

5 Conclusion and Further Works

This work presented a telemetry-based PI tuning strategy for low-level control of an OMR. We proposed the use of the MPRS method instead of PRBS to obtain the discrete-time transfer function that best reproduces each motors' behavior. It accurately represented the motor response with the real-world experiments with fitting above 90% for all tested motors while the PRBS only reached around 65% of the fitting.

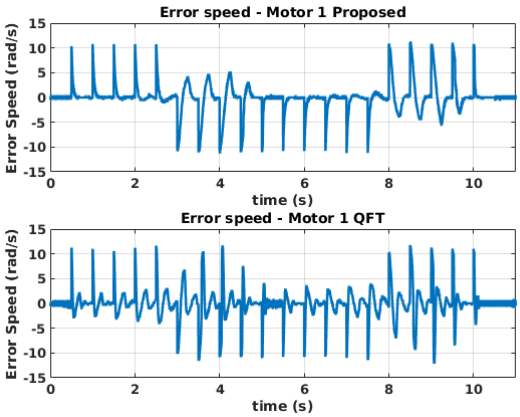


Fig. 4. Comparison of the error speed for the proposed strategy (up) and QFT approach (down).

The mechanism for data sampling based on wireless communication presented in this work allows on-site fast PI tuning to deliver suitable constants that meet the requirements of SSL competition robots. Finally, the PI controller validated in the case study had its performance compared to the QFT approach. The QFT approach's main drawbacks were the time consumed to estimate the constants, the need for manually adjusting the open-loop frequency response, and the presence of overshoot during the validation. In contrast, the proposed PI tuning strategy supports finding suitable control constants by configuring response time and transient response following motor restrictions and design requirements, reducing the testing time significantly.

The proposed strategy supports the tuning of a PI controller and tests different control constants multiple times without changing the embedded code, reducing the implementation time significantly. We can also use this strategy in a multi-robot scenario for tuning each robot's motor. Further works include validating this strategy in other types of mobile robots and implementing the telemetry-based strategy with multi-objective optimizations using the obtained control constants as initial input for evolutionary algorithms or neural networks.

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